

Module 2: Epidemic Modeling

Team Members:

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Project Title:

Modeling Disease Spread With SEIR

Project Goal:

This project seeks to analyze data of an unknown disease using exponential growth fitting and then SEIR modeling, which will then aid in the identification of the disease throughout scheduled data releases as well as possible intervention strategies.

Generative AI Statement:

Generative AI was used as assistance in understanding the properties behind SEIR modeling, how to apply Euler's method using python, and making professional plots.

1. Data and disease background

The dataset represents the spread of a mystery infectious disease within a population of approximately 17,900 individuals, which we assume corresponds to the relevant population being modeled. The dataset provides daily counts of reported active infections within the population, allowing us to examine incidence trends over time during the outbreak. While the dataset does not provide complete information needed to determine true population prevalence, the observed case counts allow us to analyze the growth of infections and estimate key epidemiological parameters such as the basic reproduction number R_0 . By examining how reported infections change over time, we can model the transmission dynamics of the disease and evaluate potential intervention strategies.

The early outbreak data shows a gradual increase in reported active infections followed by rapid exponential growth, which is typical of infectious disease outbreaks when most individuals are still susceptible. According to the SEIR model prediction, the epidemic reached a peak of approximately 3,294 active cases around day 83, meaning roughly 18–22% of the population was infected simultaneously at the peak.

The economic burden of a viral outbreak can manifest as healthcare cost, institutional expenses, and the loss of productivity on the economy as those infected miss work or school and their employers cover their sick days.

The symptoms are consistent with most upper respiratory viruses, including fever, cough, fatigue, shortness of breath, body aches, and sore throat. Respiratory viruses generally infect the respiratory tract, particularly the epithelial cells lining the nose, throat, and lungs. At the cellular level, the virus typically enters host cells by binding to receptors on the cell membrane. Once inside, the virus uses the host cell's molecular machinery to replicate its genetic material and produce viral proteins. Newly formed viral particles are then released and infect nearby cells. At the organ physiology level, infection can trigger an immune response, causing inflammation in the respiratory tract. This inflammation leads to symptoms such as coughing, mucus production, and fever. The immune system eventually clears the infection, after which individuals move into the recovered (R) category in the SEIR model.

2. Data Analysis

2a. Methods

To analyze the spread of the mystery disease, we used epidemiological data analysis and mathematical modeling techniques implemented in Python.

First, we visualized the early outbreak data by plotting reported active infections versus time. This allowed us to observe the initial growth pattern of the epidemic. Because infectious diseases often grow exponentially in the early phase of an outbreak, we fitted an exponential growth curve to the early infection data using a nonlinear least squares fitting method. From the fitted growth rate, we estimated the basic reproduction number (R_0) using the relationship between the growth rate and the recovery rate.

Next, we implemented the SEIR epidemiological model, which divides the population into four compartments:

S = Susceptible

E = Exposed

I = Infected

R = Recovered

The SEIR model is described by a system of differential equations representing the rates of transition between these compartments.

Because these equations cannot be easily solved analytically, we used Euler's method, a numerical technique that approximates solutions by updating the state of the system in

small time steps.

To determine realistic model parameters, we performed a grid search optimization over possible values of the transmission rate (β), incubation rate (σ), and recovery rate (γ). For each parameter combination, we simulated the SEIR model and compared the predicted infection curve to the observed data using the sum of squared errors (SSE). The parameter set that minimized the SSE was selected as the best-fit model.

Finally, the optimized model was used to predict the future course of the epidemic, including the timing and magnitude of the peak number of infections, as well as then applying interventions to evaluate how they affected the disease progression.

ANALYSIS AFTER DATA RELEASE #1

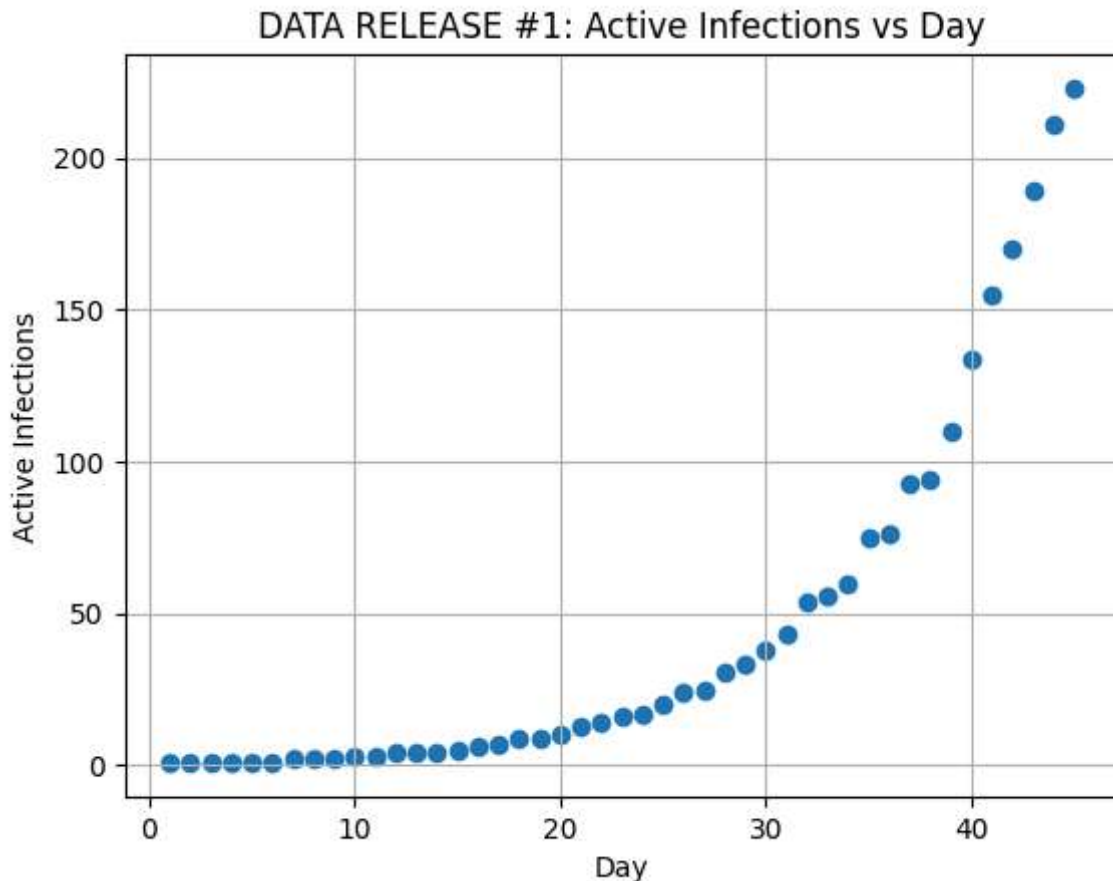
2b. Plot the data & estimate initial growth rate (R_0) from early data (through day 45)

Data release 1 tracks the active infections through day 45 of the mystery virus outbreak. The active infections per day appears to increase exponentially, which is appropriate for the first portion of an infectious disease outbreak.

```
In [ ]: import pandas as pd
import matplotlib.pyplot as plt

# Load the data
data = pd.read_csv('C:\\Users\\kiran\\Desktop\\UVA\\Classes\\Semester 4\\Computatio

# Plot
plt.figure() # Create a new figure
plt.scatter(data["day"], data["active reported daily cases"]) # Plot active infecti
plt.xlabel("Day") # Label for x-axis
plt.ylabel("Active Infections") # Label for y-axis
plt.title("DATA RELEASE #1: Active Infections vs Day") # Title for the plot
plt.grid(True) # Add a grid for better visibility
plt.show()
```



1. Regarding the initial infections, there is a slow but steady increasing slope, which then begins to increase more rapidly. This follows an exponential growth model which is characteristic of an outbreak where much of the population is at risk of infection.
2. The speed of the infection can be measured by the growth of the infected population in the early stages. By fitting an this curve with an exponential equation we can estimate the initial reproduction value, R_0 .
3. Information that would help determine the shape of the outbreak curve includes the transmission rate, incubation period, transmission rate, and the time of recovery. Furthermore, information such as contact patterns, disease interventions such as vaccines or quarantining would also impact the shape of the curve.

```
In [ ]: import pandas as pd # import pandas for data manipulation
import matplotlib.pyplot as plt # import matplotlib for plotting
import numpy as np # import numpy for numerical operations
from scipy.optimize import curve_fit # import curve_fit for fitting the exponential

# Load the data
data = pd.read_csv('C:\\Users\\kiran\\Desktop\\UVA\\Classes\\Semester 4\\Computatio

# Use the day number and active cases to fit an exponential growth curve to estimat
t = data['day'].values # time variable (day number)
I = data['active reported daily cases'].values # active infections variable (number

# Define the exponential growth function
```

```

def exponential_growth(t, r):
    return np.exp(r * t)

# Fit the exponential growth model to the data
# Initial infected population  $I_0 = 1$ 

initial_guess = [0.1] # initial guess for r
params, covariance = curve_fit(exponential_growth, t, I, p0=initial_guess) # Fit the
r_est = params[0] # Extract the estimated growth rate r from the fitted parameters

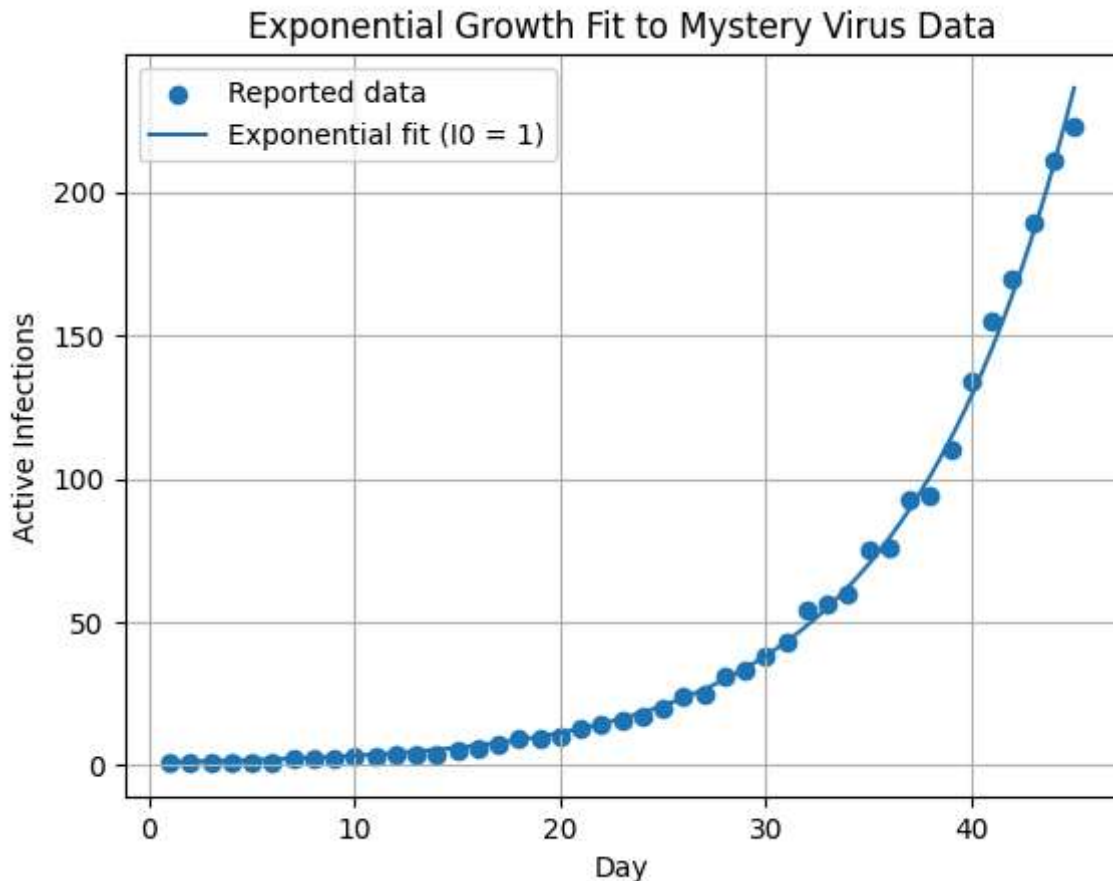
# Approximate  $R_0$  using this curve fit
gamma = 1 / 9 # recovery rate based on 9-day infectious period
R0_est = 1 + r_est / gamma #  $R_0$  can be approximated as  $1 + (\text{growth rate} / \text{recovery})$ 

# Add the fit as a line on top of scatterplot
t_fit = np.linspace(min(t), max(t), 200) # Generate a range of time values for plot
I_fit = exponential_growth(t_fit, r_est) # Calculate the fitted values using the es

plt.figure()
plt.scatter(t, I, label='Reported data') # Plot the original data points
plt.plot(t_fit, I_fit, label='Exponential fit ( $I_0 = 1$ )') # Plot the fitted exponential
plt.xlabel('Day') # Label for x-axis
plt.ylabel('Active Infections') # Label for y-axis
plt.title('Exponential Growth Fit to Mystery Virus Data') # Title for the plot
plt.legend() # Add a legend to differentiate between the data points and the fitted
plt.grid(True) # Add a grid for better visibility
plt.show()

print(f"Estimated growth rate r = {r_est:.4f} per day")
print(f"Estimated  $R_0$  = {R0_est:.2f}")

```



Estimated growth rate $r = 0.1214$ per day

Estimated $R_0 = 2.09$

4. We estimated R_0 by fitting an exponential curve to the disease growth curve. The early slope of the exponential curve approximates the reproduction number. Using the continuous growth formula with an infectious period of 9 days, the estimated growth rate was about 0.1214, and the estimated R_0 value was 2.09.
5. Our estimated R_0 value of 2.09 is similar to Ebola and Mpox, both of which have approximate R_0 values of 2. Ebola spreads via contact with bodily fluids. While this makes the virus less transmissible, it is still incredibly dangerous with a high mortality rate. Mpox is much more transmissible, being spread by close contact and respiratory droplets. Unlike Ebola, Mpox is less dangerous once transmitted, but is susceptible to outbreak.
6. The estimated $R_0=2.09$ is a reasonable, but not certain to be accurate approximation because it was obtained by fitting an exponential curve to the early disease growth phase, when transmission is least affected by immunity or interventions. However, the estimate assumes a constant growth rate, a fixed infectious period of 9 days, and homogeneous mixing, which are simplifications. As a result, the true R_0 may differ somewhat from this value, but it is likely within the correct range of moderate transmissibility.

This early estimate of R_0 is based only on the first 45 days of outbreak data, when the infections are still in the exponential growth phase. At this stage, the estimate is not very

certain because small fluctuations in early case counts can significantly alter the slope of the exponential fit model. As more data becomes available, these early estimates can be refined using more complex epidemiological models, such as SEIR. Later data allows for estimation of the incubation rate and the recovery rate, which cannot be confidently determined from early data alone.

At this stage, only the exponential growth rate is identifiable from the data, while other epidemiological parameters cannot be accurately estimated or calculated, justifying the use of exponential fitting over more complex models.

ANALYSIS AFTER DATA RELEASE #2

2c. Use Euler's method to solve the SEIR model.

Data release 2 tracks the active infections through day 70 of the mystery virus outbreak. The active infections per day the active infections continue to increase, but the growth begins to deviate from purely exponential behavior, allowing us to use the data to approximate a disease curve using Euler's method and implementing an optimized grid search to find the best fit parameters for the transmission rate, incubation rate, and recovery rate with the lowest sum of squared error. Euler's method is a first-order numerical approximation, meaning the accuracy depends on the step size. In this model we use a step size of one day, which is appropriate because the epidemiological data is reported daily (although a smaller step size would produce more accurate approximations).

Unlike the previous dataset, the disease parameters are now identifiable because the curvature of the infection data reflects transitions between susceptible, exposed, infected, and recovered compartments, allowing the disease parameters to be estimated from the data

```
In [55]: import pandas as pd # import pandas for data manipulation
import matplotlib.pyplot as plt # import matplotlib for plotting
import numpy as np # import numpy for numerical operations

# Load dataset
data = pd.read_csv('C:\\Users\\kiran\\Desktop\\UVA\\Classes\\Semester 4\\Computatio
```

```
In [56]: def run_SEIR(beta, sigma, gamma, S0, E0, I0, R0, timepoints, N): # Function to run

    # Initialize S, E, I, and R arrays
    S = np.zeros(len(timepoints))
    E = np.zeros(len(timepoints))
    I = np.zeros(len(timepoints))
    R = np.zeros(len(timepoints))

    # Set initial values
    S[0] = S0
    E[0] = E0
```

```

I[0] = I0
R[0] = R0

dt = 1 # timestep (1 day)

# Euler method loop
for t in range(len(timepoints) - 1):

    # Calculate derivatives
    dS = -beta * S[t] * I[t] / N
    dE = beta * S[t] * I[t] / N - sigma * E[t]
    dI = sigma * E[t] - gamma * I[t]
    dR = gamma * I[t]

    # Euler updates
    S[t+1] = S[t] + dS * dt
    E[t+1] = E[t] + dE * dt
    I[t+1] = I[t] + dI * dt
    R[t+1] = R[t] + dR * dt

return S, E, I, R # Return the arrays of S, E, I, and R over time

```

To model the spread of the virus, we implemented the SEIR epidemiological model, which divides the population into four compartments: S(t): Susceptible individuals who can become infected E(t): Exposed individuals who are not yet infectious I(t): Infectious individuals who can spread the disease R(t): Removed individuals who have recovered and are no longer infectious The total population, N, is assumed to remain constant. The model is governed by the following differential equations:

$$dS/dt = (-\beta)(SI)/N$$

$$dE/dt = (\beta)(SI)/N - (\sigma)(E)$$

$$dI/dt = (\sigma)(E) - (\gamma)(I)$$

$$dR/dt = (\gamma)(I)$$

Where beta is the transmission rate, sigma is the rate at which exposed individuals become infections, and gamma is the recovery rate

Because these differential equations cannot be easily solved analytically, we approximate the solution using Euler's Method, which calculates the future value by adding the current value to the derivative multiplied by the next time step size. The simulation begins with initial conditions:

S0 = initial susceptible population

E0 = initial exposed population

I0 = initial infected population

R_0 = initial recovered population

It then iterates through each time value and updates the respective values of S, E, I, and R to estimate their function values.

2d. Fit the SEIR model to the data by changing beta, gamma, and sigma.

This section should come from your python code after Data Release #2.

```
In [57]: # Function to calculate the sum of squared errors between model predictions and data
def calculate_SSE(model_I, data_I):

    n = min(len(model_I), len(data_I)) # Use the minimum length to avoid index error

    error = model_I[:n] - data_I[:n] # Calculate the error between model prediction
    SSE = np.sum(error**2) # Calculate the sum of squared errors

    return SSE # Return the calculated sum of squared errors

# Grid search to find best-fitting SEIR parameters
def fit_SEIR_gridsearch(timepoints, N, S0, E0, I0, R0, data):

    data_I = data['active reported daily cases'].values # Extract the active infect

    # Parameter ranges (centered around  $R_0 \approx 2.09$  estimate)
    beta_range = np.linspace(0.22, 0.33, 25)
    sigma_range = np.linspace(0.20, 0.35, 25)
    gamma_range = np.linspace(0.11, 0.15, 25)

    best_SSE = np.inf # Initialize best SSE to infinity for comparison
    best_beta = None # Initialize best beta to None
    best_sigma = None # Initialize best sigma to None
    best_gamma = None # Initialize best gamma to None

    for b in beta_range: # Iterate over the range of beta values
        for s in sigma_range: # Iterate over the range of sigma values
            for g in gamma_range: # Iterate over the range of gamma values

                S, E, I, R = run_SEIR(b, s, g, S0, E0, I0, R0, timepoints, N) # Run

                SSE = calculate_SSE(I, data_I) # Calculate the sum of squared error

                if SSE < best_SSE: # If the current SSE is better than the best SSE
                    best_SSE = SSE
                    best_beta = b
                    best_sigma = s
                    best_gamma = g

    return best_beta, best_sigma, best_gamma, best_SSE

# Set initial conditions and run grid search to fit SEIR model to data
```

```

# Population information
N = 17900 # Total population size (assumed for the region being modeled)

I0 = 1 # Initial number of infected individuals (starting with 1 infected person)
E0 = 5 # Initial number of exposed individuals (assumed to be 5 based on early grow
R0 = 0 # Initial number of recovered individuals (starting with 0 recovered)
S0 = N - I0 - E0 - R0 # Initial number of susceptible individuals (the rest of the

# Timepoints matching dataset
timepoints = np.arange(0, len(data), 1)

# Run grid search
best_beta, best_sigma, best_gamma, best_SSE = fit_SEIR_gridsearch(timepoints, N, S0

print("Best beta:", best_beta) # Print the best beta value found from the grid sear
print("Best sigma:", best_sigma) # Print the best sigma value found from the grid s
print("Best gamma:", best_gamma) # Print the best gamma value found from the grid s
print("Best SSE:", best_SSE) # Print the best sum of squared errors corresponding t

print("Estimated R0 from model:", best_beta / best_gamma) # Calculate and print the

```

```

Best beta: 0.33
Best sigma: 0.24375
Best gamma: 0.11333333333333333
Best SSE: 120826.14686809502
Estimated R0 from model: 2.9117647058823533

```

To determine realistic values for beta, sigma, and gamma, we used a grid search optimization approach that tested multiple combinations of these parameters. The ranges for beta, sigma, and gamma, were selected using the early estimate of $R_0 = 2.09$. Because $R_0 = \beta/\gamma$, we selected beta and gamma ranges that would produce values of R_0 close to this estimate. The sigma range was chosen to represent the latent periods of roughly 3-5 days, which are typical for respiratory viral infections. By restricting the grid search optimization to biologically plausible ranges informed by the initial R_0 estimate and known disease characteristics, the optimization avoids unrealistic parameter combinations while improving efficiency towards finding a logical solution.

For each combination, the SEIR model was simulated using Euler's method, the predicted infected population over time was compared to the actual infected data, and the sum of squared error (SSE) was calculated. The parameters that gave the lowest SSE were selected as the best fit for the dataset. In this case, the grid search optimization found the best beta, sigma, and gamma values to be 0.33, 0.24375, and 0.113, respectively, which a best SSE of 120826.

Additionally, differences between the initial exponential estimate of R_0 (2.09) and the SEIR-optimized value (2.91) highlight how early outbreak data can underestimate transmission once full disease dynamics are considered.

2e. Plot the model-predicted infections over time compared to the data.

```
In [58]: # We extend the simulation to 300 days so the epidemic has time to reach its peak
long_timepoints = np.arange(0, 300, 1)

# Run the SEIR model using the best parameters found from the grid search
# This returns arrays representing the number of Susceptible, Exposed,
# Infectious, and Recovered individuals at each time point
S_long, E_long, I_long, R_long = run_SEIR(
    best_beta,      # transmission rate
    best_sigma,     # rate exposed individuals become infectious
    best_gamma,     # recovery rate
    S0,             # initial susceptible population
    E0,             # initial exposed population
    I0,             # initial infected population
    R0,             # initial recovered population
    long_timepoints, # timepoints for the simulation
    N               # total population size
)

# Create a new figure
plt.figure()

# Plot the reported infection data from the dataset
plt.scatter(timepoints, data['active reported daily cases'], label="Reported Data")

# Plot the SEIR model prediction over the longer time range
plt.plot(long_timepoints, I_long, label="SEIR Model Prediction")

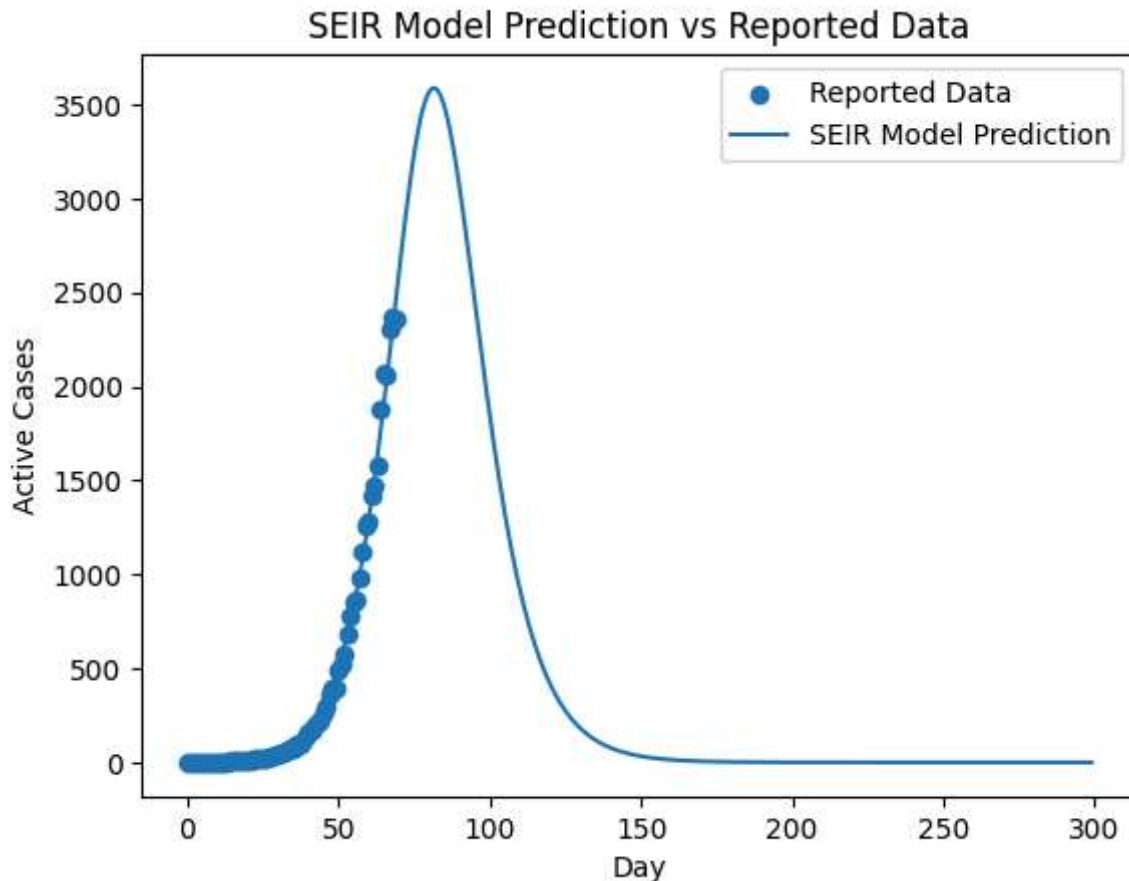
# Label the x-axis
plt.xlabel("Day")

# Label the y-axis
plt.ylabel("Active Cases")

# Add a title to describe the plot
plt.title("SEIR Model Prediction vs Reported Data")

# Add a legend to distinguish between data and model
plt.legend()

# Show the plot
plt.show()
```



This code identifies the peak of the epidemic predicted by the SEIR model. The `np.max()` function finds the largest value in the predicted infection array `I_long`, which represents the highest number of active cases. The `np.argmax()` function finds the day when this maximum value occurs. Finally, the code calculates the percentage of the total population infected at the peak by dividing the peak cases by the total population size.

2f. Predict the day and amount of active cases at the peak of the epidemic spread

```
In [59]: ### 2e. Predict the day and amount of active cases at the peak of the epidemic spread

#Find the maximum number of infections predicted by the model
np.max # returns the largest value in the I_long array
peak_cases = np.max(I_long)

#Find the day when that maximum value occurs
np.argmax # returns the index of the largest value
peak_day = np.argmax(I_long)
# Print the predicted peak number of active infections
print("Predicted peak active cases:", peak_cases)

# Print the day when the epidemic reaches its peak
print("Predicted peak day:", peak_day)

# Calculate the percentage of the total population infected at the peak
```

```
percent_population = (peak_cases / N) * 100

# Print the percent of the population infected at the peak
print("Percent of population infected at peak:", percent_population)
```

Predicted peak active cases: 3586.3941647270526

Predicted peak day: 82

Percent of population infected at peak: 20.035721590653928

The SEIR model predicts that the epidemic will reach a peak of approximately 3586 active cases. This peak is expected to occur on day 82 of the outbreak. At this point, about 20% of the total population is infected simultaneously. This value is reasonable given the population size of 17,900 and the rapid growth observed in the reported case data up until day 70.

While these optimized parameters provide a strong fit to the observed data, some uncertainty remains because the dataset does not fully capture long-term dynamics or potential changes in population behavior, meaning multiple parameter combinations could produce similarly good fits.

ANALYSIS AFTER DATA RELEASE #3

2g. Plot the full dataset (Data Release #3) against your model.

Data release 3 tracks the active infections through day 120 of the mystery virus outbreak. The active infections per day reach a peak, and the SEIR optimized model was modeled against the final data release. At this stage, the full dynamics of the outbreaks are observable, making it possible to evaluate how well the SEIR optimized model predicted the disease behavior.

```
In [60]: import pandas as pd # import pandas for data manipulation
import matplotlib.pyplot as plt # import matplotlib for plotting
import numpy as np # import numpy for numerical operations

# Load dataset
data = pd.read_csv('C:\\Users\\kiran\\Desktop\\UVA\\Classes\\Semester 4\\Computatio
```

```
In [61]: plt.figure()

# Plot Release 3 data
plt.scatter(
    data['day'], # x-axis: day number from the dataset
    data['active reported daily cases'], # y-axis: active cases from the dataset
    label="Release #3 Data" # Label for the scatter points representing the data fr
)

# Plot SEIR model prediction
plt.plot(
    long_timepoints, # x-axis: time points for the model prediction
    I_long, # y-axis: predicted active cases from the SEIR model
```

```

    label="SEIR Model Prediction" # Label for the line representing the model prediction
)

plt.xlabel("Day") # Label for x-axis
plt.ylabel("Active Cases") # Label for y-axis
plt.title("SEIR Model vs Data Release #3") # Title for the plot comparing the SEIR
plt.legend()

plt.show()

# True peak values from the dataset
true_peak_cases = np.max(data['active reported daily cases']) # Find the maximum number of cases
true_peak_day = data['day'][data['active reported daily cases'].idxmax()] # Find the day of the peak

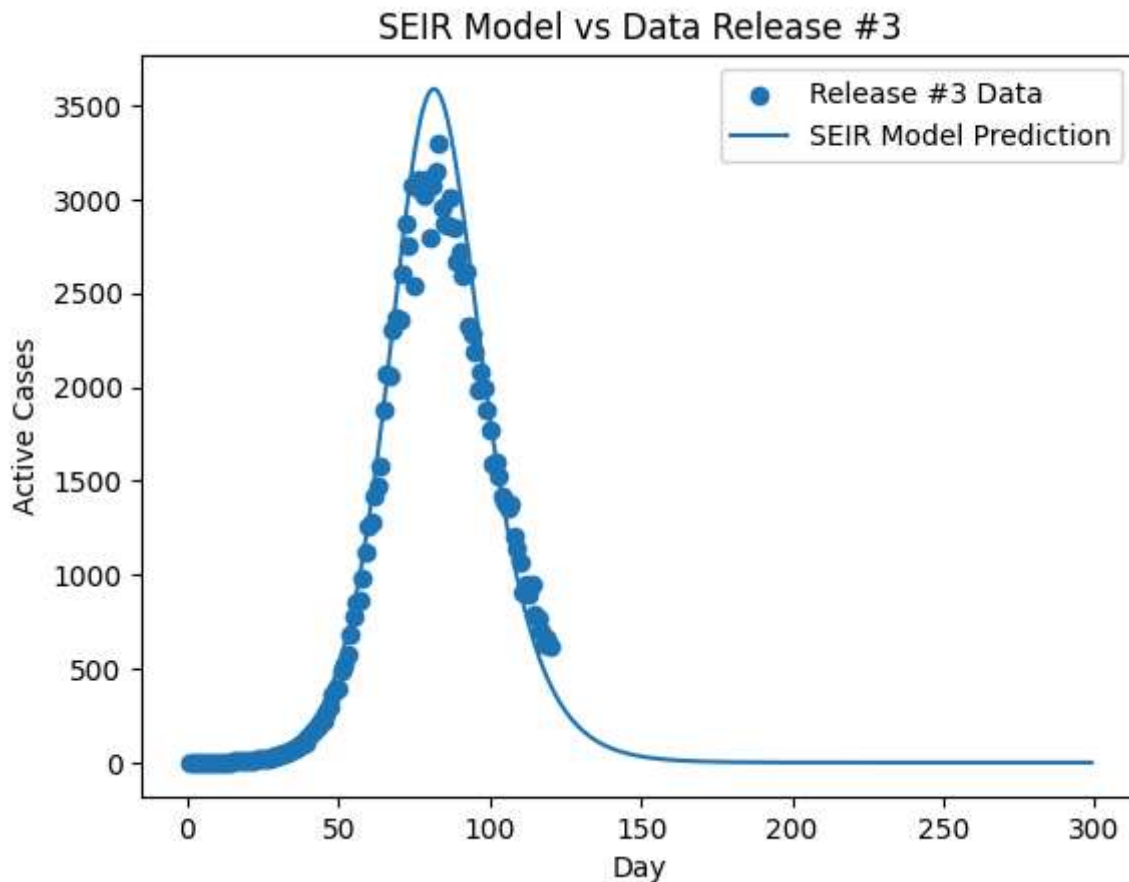
print("True peak cases (Release 3):", true_peak_cases)
print("True peak day (Release 3):", true_peak_day)

# Percent relative error for peak cases (y error)
error_cases = abs(true_peak_cases - peak_cases) / true_peak_cases * 100

# Percent relative error for peak day (x error)
error_day = abs(true_peak_day - peak_day) / true_peak_day * 100

print("Percent relative error (peak cases):", error_cases)
print("Percent relative error (peak day):", error_day)

```



True peak cases (Release 3): 3294
True peak day (Release 3): 83
Percent relative error (peak cases): 8.876568449515867
Percent relative error (peak day): 1.2048192771084338

Using our model, we predicted that the peak number of cases would be 3586. Upon receiving the expanded data, we found the true peak to be 3294 cases, which is 8.88% lower than we predicted. Using our earlier models, we predicted the day with the peak number of cases to be 82, which was 1.2% off from the true number of day 83. Overall, these errors are relatively low given the limited data available, and the model predicted the peak day and cases relatively accurately.

There are a few possible causes for this error. In real epidemics, the transmission rate, incubation rate, and recovery rate change as the disease develops and new strains emerge. However, in our model, we assigned each of these values as constant. Second, our code did not take into account the behavior of the affected individuals. In reality, individuals may resist isolation, vaccines, and preventative measures. Our code treated each case as though the infected individuals. We also assumed homogenous mixing, which would mean that everyone in the population would have an equal risk of getting infected. However, that is not always accurate. Last, our model was made using the early outbreak data, which models an exponential curve. However, smaller errors in the early parameter estimates can produce larger errors later in the epidemic when the model is projected forward, which likely contributed to the overestimation of the peak number of cases.

Despite these limitations, the low error in both peak cases and the day of the peak suggest that the model is still effective at approximating overall epidemic trends, making it a useful tool for comparing intervention strategies and informing public health decisions.

Exponential growth fitting worked early when not many data points were known, but failed after more time passed due to it assuming unlimited susceptible individuals, which is not the case in a true disease outbreak. As the outbreak progresses, the number of susceptible people decreases and the number of infections begin to decrease. An exponential growth model cannot accurately describe those dynamics. Therefore, SEIR modelling becomes necessary to estimate the progression of the disease after additional data has been released.

2h. Intervention strategies for new outbreak at VT (70 days of infection)

This section should come from your python code after Data Release #3.

Baseline at Virginia Tech

The disease model using the best beta, sigma, and gamma was applied to Virginia Tech's student population (38,000 students) to model outbreak patterns in Blacksburg and how different interventions would reduce the number of infections over a period of time. Each

intervention was implemented by modifying specific SEIR model parameters, allowing us to directly evaluate how changes in transmission dynamics affect epidemic outcomes.

```
In [62]: # Virginia Tech population
N_VT = 38000

# Initial conditions
I0_VT = 1
E0_VT = 5
R0_VT = 0
S0_VT = N_VT - I0_VT - E0_VT - R0_VT

# Timepoints for longer simulation
timepoints_VT = np.arange(0, 300, 1)

# Run SEIR model with same optimized parameters
S_VT, E_VT, I_VT, R_VT = run_SEIR(
    best_beta,
    best_sigma,
    best_gamma,
    S0_VT,
    E0_VT,
    I0_VT,
    R0_VT,
    timepoints_VT,
    N_VT
)

# Plot the predicted active cases for the Virginia Tech population
plt.figure()

plt.plot(timepoints_VT, I_VT, label="Predicted Active Cases (VT Population)")

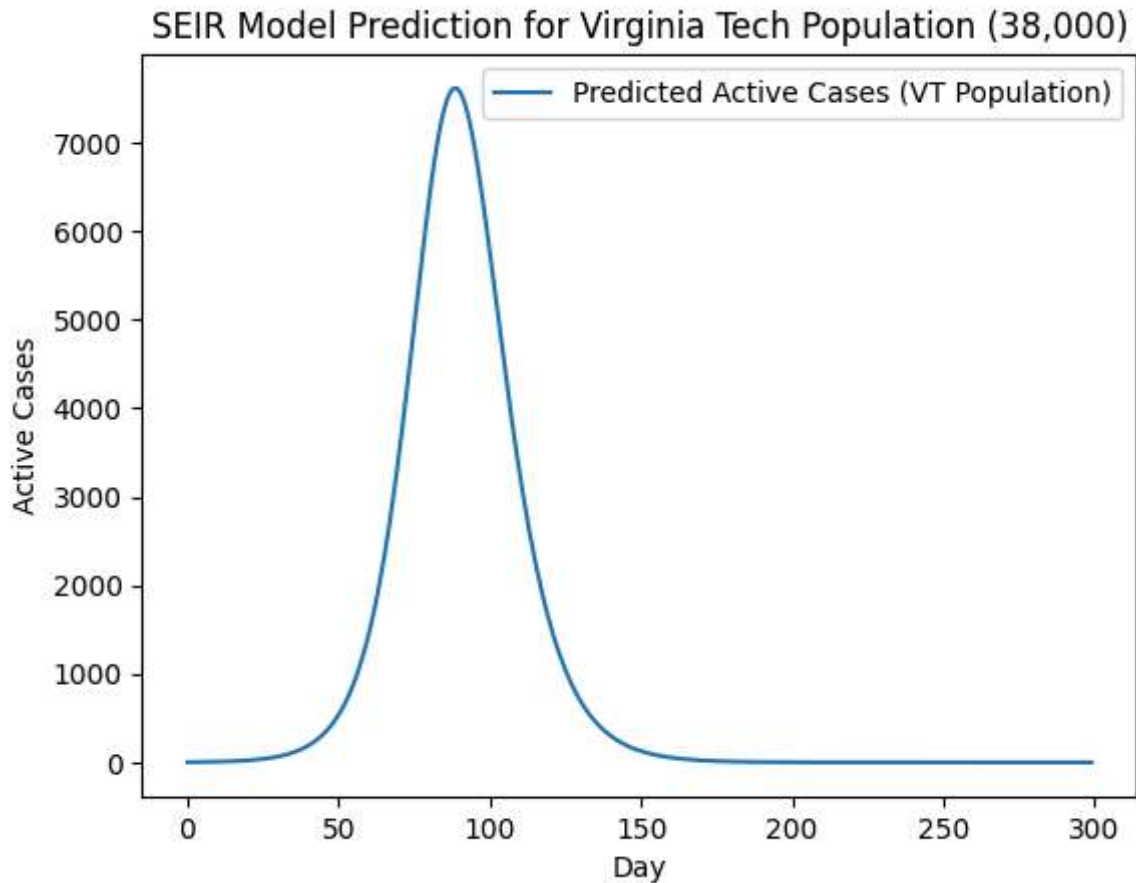
plt.xlabel("Day") # Label for x-axis
plt.ylabel("Active Cases") # Label for y-axis
plt.title("SEIR Model Prediction for Virginia Tech Population (38,000)") # Title fo
plt.legend()

plt.show()

# Peak infections and day
VT_peak_cases = np.max(I_VT) # Find the maximum number of active cases predicted fo
VT_peak_day = np.argmax(I_VT) # Find the day corresponding to the maximum number of

print("VT predicted peak cases:", VT_peak_cases)
print("VT predicted peak day:", VT_peak_day)

# Percent of population infected at peak
VT_peak_percent = (VT_peak_cases / N_VT) * 100
print("Percent of VT population infected at peak:", VT_peak_percent)
```



VT predicted peak cases: 7613.134117641253

VT predicted peak day: 89

Percent of VT population infected at peak: 20.034563467476982

At Virginia Tech, with a higher student population, the model predicted number of peak cases was much higher than the peak at the University of Virginia, at about 7613, and it occurred at day 89. However, the percent infected at peak was predicted to be about the same in both universities (20%).

The larger population at Virginia Tech increases the total number of infections predicted by the SEIR model, but the percent infected at peak remains relatively consistent because the SEIR model depends primarily on the transmission parameters (beta, sigma, gamma) than the size of the population at risk.

Intervention Strategy 1: Masking Mandates

One intervention strategy simulated is implementing mask mandates at day 70 with varying strengths: cloth masks at about 26.5%, surgical masks at about 40%, and N95 masks at about 70% reduced transmission rate (reduces the value of beta proportionally):

References:

Clapp PW, Sickbert-Bennett EE, Samet JM, et al. Evaluation of Cloth Masks and Modified Procedure Masks as Personal Protective Equipment for the Public During the COVID-19

Pandemic. JAMA Intern Med. 2021;181(4):463–469. doi:10.1001/jamainternmed.2020.8168

Floriano I, Silvinato A, Bacha HA, Barbosa AN, Tanni S, Bernardo WM. Effectiveness of wearing masks during the COVID-19 outbreak in cohort and case-control studies: a systematic review and meta-analysis. J Bras Pneumol. 2024 Jan 5;49(6):e20230003. doi: 10.36416/1806-3756/e20230003. PMID: 38198343; PMCID: PMC10760436.

Mask mandate SEIR simulation function

```
In [63]: def run_SEIR_mask(beta, sigma, gamma, S0, E0, I0, R0, timepoints, N, mask_day, mask_strength):

    S = [S0] # Initialize the list for susceptible individuals with the initial value
    E = [E0] # Initialize the list for exposed individuals with the initial value
    I = [I0] # Initialize the list for infected individuals with the initial value
    R = [R0] # Initialize the list for recovered individuals with the initial value

    for t in range(len(timepoints)-1):

        # Apply mask mandate after mask_day
        if t >= mask_day:
            beta_effective = beta * (1 - mask_strength)
        else:
            beta_effective = beta

        # SEIR derivatives
        dS = -beta_effective * S[t] * I[t] / N
        dE = beta_effective * S[t] * I[t] / N - sigma * E[t]
        dI = sigma * E[t] - gamma * I[t]
        dR = gamma * I[t]

        # Euler update
        S.append(S[t] + dS)
        E.append(E[t] + dE)
        I.append(I[t] + dI)
        R.append(R[t] + dR)

    return np.array(S), np.array(E), np.array(I), np.array(R)
```

Set VT population

```
In [64]: N_VT = 38000 # Total population of Virginia Tech

I0_VT = 1 # Initial infected individuals at VT
E0_VT = 5 # Initial exposed individuals at VT
R0_VT = 0 # Initial recovered individuals at VT
S0_VT = N_VT - I0_VT - E0_VT - R0_VT # Initial susceptible individuals at VT

timepoints_VT = np.arange(0, 200, 1)
```

Run baseline and mask mandate scenarios

```

In [65]: S_base, E_base, I_base, R_base = run_SEIR(
    best_beta,
    best_sigma,
    best_gamma,
    S0_VT,
    E0_VT,
    I0_VT,
    R0_VT,
    timepoints_VT,
    N_VT
) # Run the baseline SEIR model without masks for the VT population

mask_day = 70 # Assume mask mandate starts on day 70 of the simulation

# Mask strengths
mask_265 = 0.265 # Mask strength for cloth masks (26.5% reduction in transmission)
mask_40 = 0.40 # Mask strength for surgical masks (40% reduction in transmission)
mask_70 = 0.70 # Mask strength for N95 masks (70% reduction in transmission)

S_265, E_265, I_265, R_265 = run_SEIR_mask(best_beta, best_sigma, best_gamma,
    S0_VT, E0_VT, I0_VT, R0_VT,
    timepoints_VT, N_VT,
    mask_day, mask_265) # Run the SEIR model

S_40, E_40, I_40, R_40 = run_SEIR_mask(best_beta, best_sigma, best_gamma,
    S0_VT, E0_VT, I0_VT, R0_VT,
    timepoints_VT, N_VT,
    mask_day, mask_40) # Run the SEIR model with

S_70, E_70, I_70, R_70 = run_SEIR_mask(best_beta, best_sigma, best_gamma,
    S0_VT, E0_VT, I0_VT, R0_VT,
    timepoints_VT, N_VT,
    mask_day, mask_70) # Run the SEIR model with

```

Compare baseline to mask scenarios

```

In [66]: plt.figure()

plt.plot(timepoints_VT, I_base, label="Baseline (No Masks)") # Plot the baseline SE
plt.plot(timepoints_VT, I_265, label="Masking 26.5% Reduction") # Plot the SEIR mod
plt.plot(timepoints_VT, I_40, label="Masking 40% Reduction") # Plot the SEIR model
plt.plot(timepoints_VT, I_70, label="Masking 70% Reduction") # Plot the SEIR model

plt.axvline(mask_day, linestyle='--', label="Mask Mandate Start (Day 70)") # Add a

plt.xlabel("Day") # Label for x-axis
plt.ylabel("Active Infections") # Label for y-axis
plt.title("Impact of Mask Mandate on VT Outbreak") # Title for the plot showing the
plt.legend()

plt.show()

# Baseline
baseline_peak_cases = np.max(I_base) # Find the maximum number of active cases pred

```

```

baseline_peak_day = np.argmax(I_base) # Find the day corresponding to the maximum n
# 26.5% masking
peak_265_cases = np.max(I_265) # Find the maximum number of active cases predicted
peak_265_day = np.argmax(I_265) # Find the day corresponding to the maximum number
# 40% masking
peak_40_cases = np.max(I_40) # Find the maximum number of active cases predicted fo
peak_40_day = np.argmax(I_40) # Find the day corresponding to the maximum number of
# 70% masking
peak_70_cases = np.max(I_70) # Find the maximum number of active cases predicted fo
peak_70_day = np.argmax(I_70) # Find the day corresponding to the maximum number of

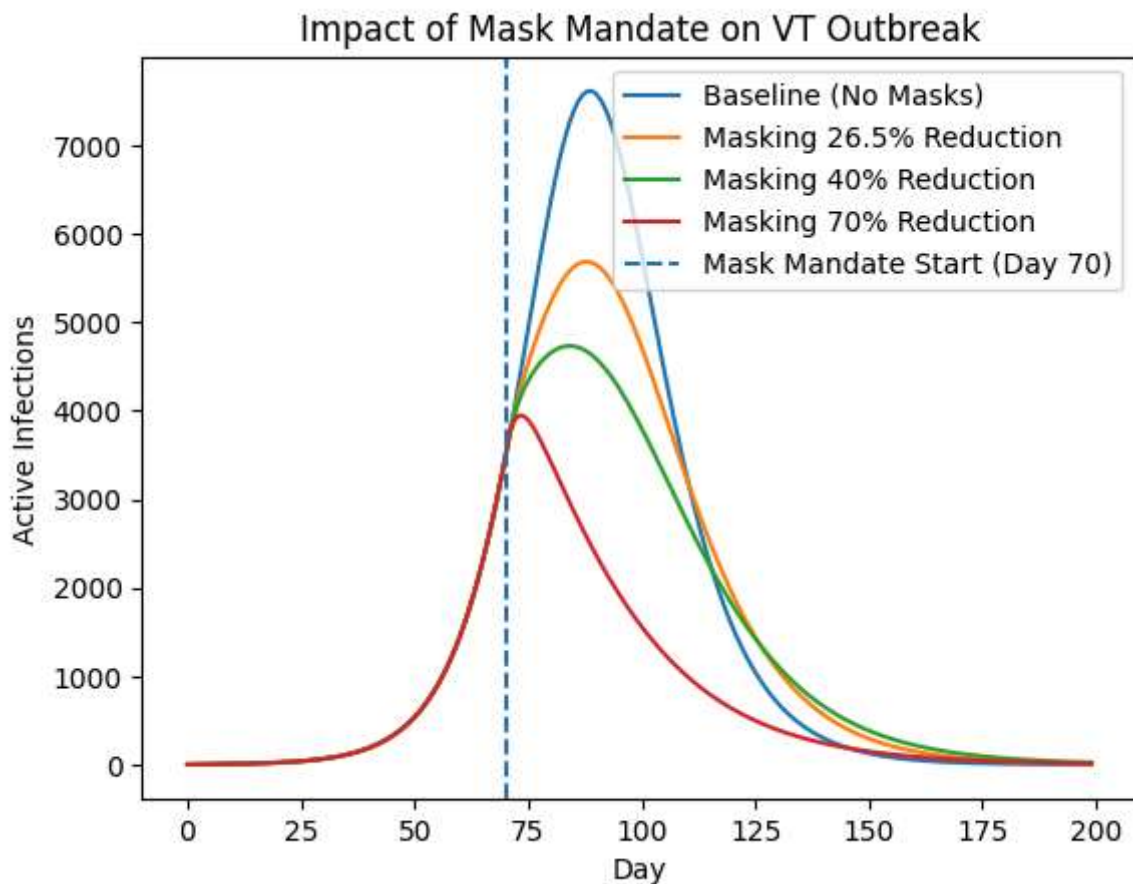
print("Baseline peak:", baseline_peak_cases, "on day", baseline_peak_day)

print("26.5% masking peak:", peak_265_cases, "on day", peak_265_day)

print("40% masking peak:", peak_40_cases, "on day", peak_40_day)

print("70% masking peak:", peak_70_cases, "on day", peak_70_day)

```



Baseline peak: 7613.134117641253 on day 89
 26.5% masking peak: 5684.384559563923 on day 88
 40% masking peak: 4732.400169307035 on day 84
 70% masking peak: 3947.4743592487976 on day 73

Results

Implementing mask mandates at day 70 was a successful method of reducing peak cases. It also reduced the day the number of peak cases occurred. As expected, the highest performing mask (N95) reduced cases the most compared to baseline, while the least performing masks (cloth, no filter) reduced cases the least compared to baseline.

Compared to the baseline scenario, N95 masking reduced the peak number of infections by approximately 48.15%, while surgical masks reduced the peak by approximately 37.84%. Cloth masks produced a smaller reduction of approximately 25.34%. This demonstrates that stronger reductions in transmission rate lead to significantly lower epidemic peaks.

However, this model assumes 100% compliance from the entire VT population, which as seen during the COVID-19 pandemic, is not entirely realistic.

Intervention Strategy 2: Vaccine Campaign

Another intervention strategy is a mass vaccine event, vaccinating 2000 susceptible students with 90% efficacy (effectively meaning 1800 susceptible students move to the recovered label). The effects of a vaccine event was tested across timeframes of when said event could happen: day 40 after the outbreak, day 70, and day 100.

SEIR function with vaccination event

```
In [67]: def run_SEIR_vaccine(beta, sigma, gamma, S0, E0, I0, R0, timepoints, N, vaccine_day

    S = [S0] # Initialize the list for susceptible individuals with the initial val
    E = [E0] # Initialize the list for exposed individuals with the initial value E
    I = [I0] # Initialize the list for infected individuals with the initial value
    R = [R0] # Initialize the list for recovered individuals with the initial value

    vaccinated_effective = vaccines * efficacy # Calculate the effective number of

    for t in range(len(timepoints)-1):

        S_current = S[t]
        E_current = E[t]
        I_current = I[t]
        R_current = R[t]

        # Vaccination event
        if t == vaccine_day:
            vaccinated = min(vaccinated_effective, S_current)
            S_current -= vaccinated
            R_current += vaccinated

        # SEIR derivatives
        dS = -beta * S_current * I_current / N
        dE = beta * S_current * I_current / N - sigma * E_current
        dI = sigma * E_current - gamma * I_current
        dR = gamma * I_current
```

```

    # Euler update
    S.append(S_current + dS)
    E.append(E_current + dE)
    I.append(I_current + dI)
    R.append(R_current + dR)

    return np.array(S), np.array(E), np.array(I), np.array(R)

```

Run vaccination scenarios

```

In [68]: N_VT = 38000 # Total population of Virginia Tech

I0_VT = 1 # Initial infected individuals at VT
E0_VT = 5 # Initial exposed individuals at VT
R0_VT = 0 # Initial recovered individuals at VT
S0_VT = N_VT - I0_VT - E0_VT - R0_VT # Initial susceptible individuals at VT

timepoints_VT = np.arange(0, 200, 1)

vaccines = 2000 # Number of vaccines administered on the vaccination day
efficacy = 0.90 # Efficacy of the vaccine (90% reduction in susceptibility for vacc

# Day 40 vaccination
S_v40, E_v40, I_v40, R_v40 = run_SEIR_vaccine(
    best_beta, best_sigma, best_gamma,
    S0_VT, E0_VT, I0_VT, R0_VT,
    timepoints_VT, N_VT,
    vaccine_day=40,
    vaccines=vaccines,
    efficacy=efficacy
) # Run the SEIR model with a vaccination event on day 40 for the VT population

# Day 70 vaccination
S_v70, E_v70, I_v70, R_v70 = run_SEIR_vaccine(
    best_beta, best_sigma, best_gamma,
    S0_VT, E0_VT, I0_VT, R0_VT,
    timepoints_VT, N_VT,
    vaccine_day=70,
    vaccines=vaccines,
    efficacy=efficacy
) # Run the SEIR model with a vaccination event on day 70 for the VT population

# Day 100 vaccination
S_v100, E_v100, I_v100, R_v100 = run_SEIR_vaccine(
    best_beta, best_sigma, best_gamma,
    S0_VT, E0_VT, I0_VT, R0_VT,
    timepoints_VT, N_VT,
    vaccine_day=100,
    vaccines=vaccines,
    efficacy=efficacy
) # Run the SEIR model with a vaccination event on day 100 for the VT population

```

Plot vaccination scenarios vs baseline

```

In [69]: plt.figure()

plt.plot(timepoints_VT, I_base, label="Baseline (No Vaccine)") # Plot the baseline
plt.plot(timepoints_VT, I_v40, label="Vaccine Event Day 40") # Plot the SEIR model
plt.plot(timepoints_VT, I_v70, label="Vaccine Event Day 70") # Plot the SEIR model
plt.plot(timepoints_VT, I_v100, label="Vaccine Event Day 100") # Plot the SEIR model

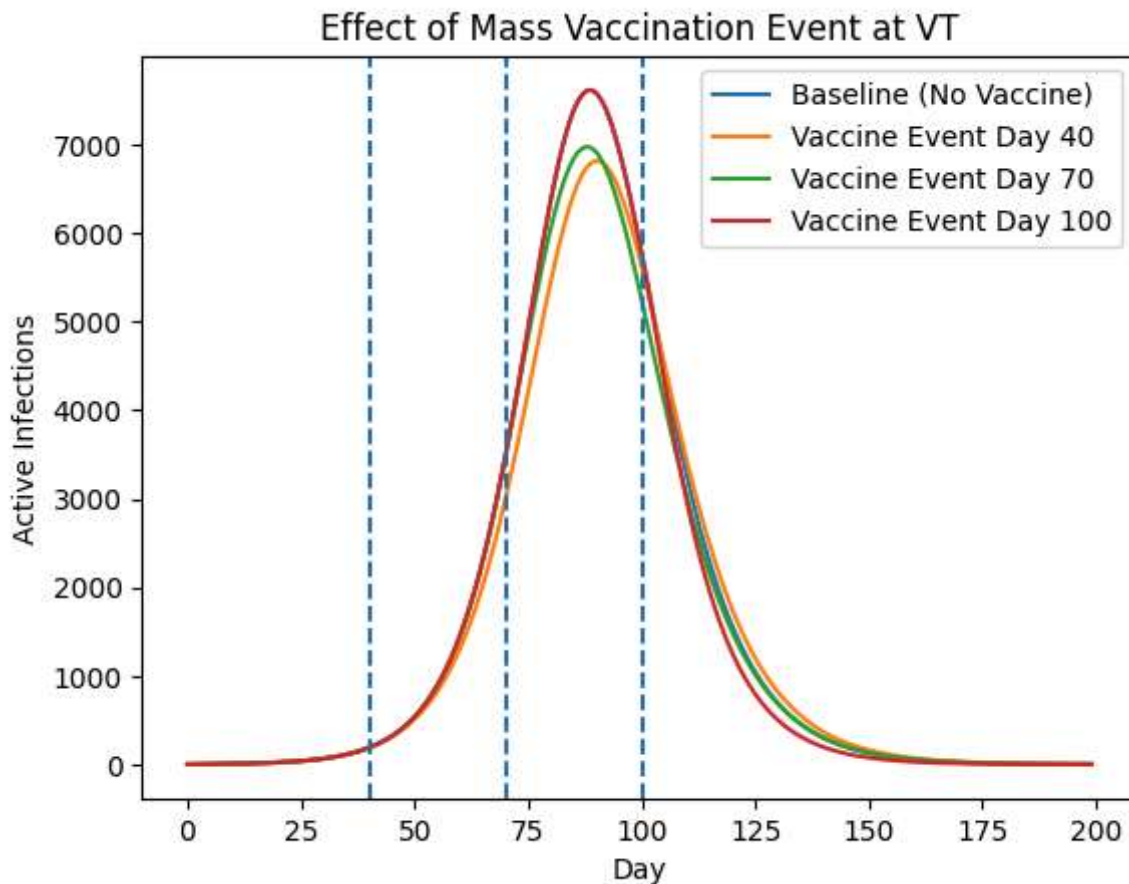
plt.axvline(40, linestyle='--') # Add a vertical dashed line to indicate when the v
plt.axvline(70, linestyle='--') # Add a vertical dashed line to indicate when the v
plt.axvline(100, linestyle='--') # Add a vertical dashed line to indicate when the

plt.xlabel("Day") # Label for x-axis
plt.ylabel("Active Infections") # Label for y-axis
plt.title("Effect of Mass Vaccination Event at VT") # Title for the plot showing th
plt.legend()

plt.show()

print("Baseline peak:", np.max(I_base), "day", np.argmax(I_base)) # Print the peak
print("Vaccine day 40 peak:", np.max(I_v40), "day", np.argmax(I_v40)) # Print the p
print("Vaccine day 70 peak:", np.max(I_v70), "day", np.argmax(I_v70)) # Print the p
print("Vaccine day 100 peak:", np.max(I_v100), "day", np.argmax(I_v100)) # Print th

```



Baseline peak: 7613.134117641253 day 89
 Vaccine day 40 peak: 6815.8304848147945 day 90
 Vaccine day 70 peak: 6975.63765547902 day 88
 Vaccine day 100 peak: 7613.134117641253 day 89

Results

Vaccination at day 40 reduced peak infections by approximately 10.47% compared to baseline, while vaccination at day 70 reduced the peak by approximately 8.37%. Vaccination at day 100 had minimal impact, not reducing the peak number of infections at all, demonstrating the importance of early intervention.

Interestingly, earlier vaccine rollout at day 40 leads to a later peak day (90) than baseline (89) and vaccine rollout at day 70 (88). However, having enough data to have a large vaccine event is ambitious by day 40, so a day 70 vaccination event is more realistic and almost as effective at reducing peak cases.

Intervention Strategy 3: School Closure

A final intervention strategy is a school closure at day 70, which reduces the contact rate by 80% (reduces beta to 20% of its original value). Different lengths (1 week, 2 weeks, 3 weeks) of closure will be tested, and then the contact rate will return to normal.

SEIR model with temporary school closure

```
In [70]: def run_SEIR_school_closure(beta, sigma, gamma, S0, E0, I0, R0, timepoints, N, clos

    S = [S0] # Initialize the list for susceptible individuals with the initial val
    E = [E0] # Initialize the list for exposed individuals with the initial value E
    I = [I0] # Initialize the list for infected individuals with the initial value
    R = [R0] # Initialize the list for recovered individuals with the initial value

    for t in range(len(timepoints)-1):

        S_current = S[t]
        E_current = E[t]
        I_current = I[t]
        R_current = R[t]

        # Adjust beta during school closure
        if closure_day <= t < closure_day + closure_length:
            beta_t = beta * (1 - reduction) # reduce beta
        else:
            beta_t = beta

        # SEIR derivatives
        dS = -beta_t * S_current * I_current / N
        dE = beta_t * S_current * I_current / N - sigma * E_current
        dI = sigma * E_current - gamma * I_current
        dR = gamma * I_current
```

```

    # Euler update
    S.append(S_current + dS)
    E.append(E_current + dE)
    I.append(I_current + dI)
    R.append(R_current + dR)

    return np.array(S), np.array(E), np.array(I), np.array(R)

```

Run closure scenarios

```

In [71]: N_VT = 38000 # Total population of Virginia Tech

I0_VT = 1 # Initial infected individuals at VT
E0_VT = 5 # Initial exposed individuals at VT
R0_VT = 0 # Initial recovered individuals at VT
S0_VT = N_VT - I0_VT - E0_VT - R0_VT # Initial susceptible individuals at VT

timepoints_VT = np.arange(0,200,1)

closure_day = 70 # Day when school closure starts
reduction = 0.80 # 80% reduction in contact rate

# 1 week closure
S_c7, E_c7, I_c7, R_c7 = run_SEIR_school_closure(
    best_beta, best_sigma, best_gamma,
    S0_VT, E0_VT, I0_VT, R0_VT,
    timepoints_VT, N_VT,
    closure_day, 7, reduction
) # Run the SEIR model with a school closure starting at day 70 for 7 days, reducin

# 2 week closure
S_c14, E_c14, I_c14, R_c14 = run_SEIR_school_closure(
    best_beta, best_sigma, best_gamma,
    S0_VT, E0_VT, I0_VT, R0_VT,
    timepoints_VT, N_VT,
    closure_day, 14, reduction
) # Run the SEIR model with a school closure starting at day 70 for 14 days, reduci

# 3 week closure
S_c21, E_c21, I_c21, R_c21 = run_SEIR_school_closure(
    best_beta, best_sigma, best_gamma,
    S0_VT, E0_VT, I0_VT, R0_VT,
    timepoints_VT, N_VT,
    closure_day, 21, reduction
) # Run the SEIR model with a school closure starting at day 70 for 21 days, reduci

```

Plot results vs baseline

```

In [72]: plt.figure()

plt.plot(timepoints_VT, I_base, label="Baseline (No Closure)") # Plot the baseline
plt.plot(timepoints_VT, I_c7, label="1 Week Closure") # Plot the SEIR model predict
plt.plot(timepoints_VT, I_c14, label="2 Week Closure") # Plot the SEIR model predic
plt.plot(timepoints_VT, I_c21, label="3 Week Closure") # Plot the SEIR model predic

```

```

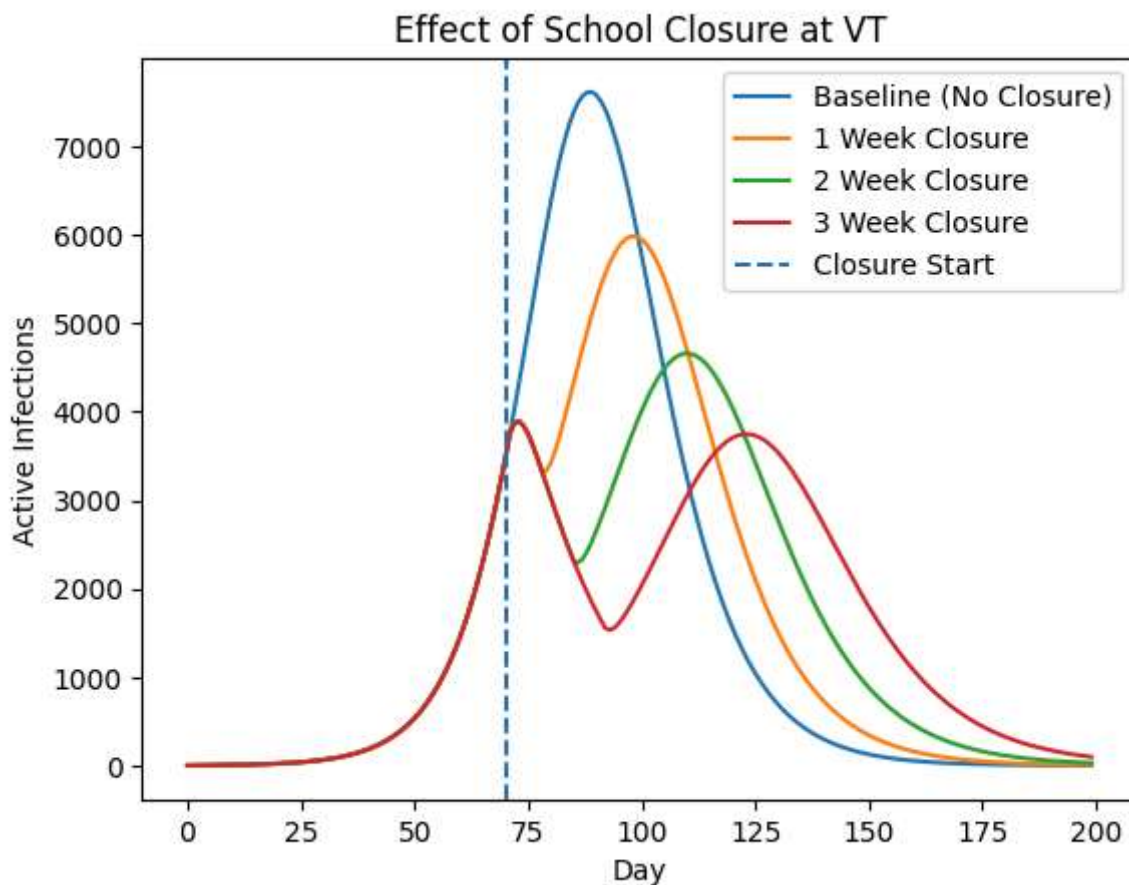
plt.axvline(70, linestyle='--', label="Closure Start") # Add a vertical dashed line

plt.xlabel("Day") # Label for x-axis
plt.ylabel("Active Infections") # Label for y-axis
plt.title("Effect of School Closure at VT") # Title for the plot showing the effect
plt.legend()

plt.show()

print("Baseline peak:", np.max(I_base), "day", np.argmax(I_base)) # Print the peak
print("1 week closure peak:", np.max(I_c7), "day", np.argmax(I_c7)) # Print the peak
print("2 week closure peak:", np.max(I_c14), "day", np.argmax(I_c14)) # Print the peak
print("3 week closure peak:", np.max(I_c21), "day", np.argmax(I_c21)) # Print the peak

```



```

Baseline peak: 7613.134117641253 day 89
1 week closure peak: 5984.588647760676 day 98
2 week closure peak: 4659.350603114125 day 110
3 week closure peak: 3892.3157786720176 day 120

```

Results

School closure for 1 week reduced peak infections by approximately 21.38% compared to baseline, 2 weeks reduced peak infections by approximately 38.80% compared to baseline, and 3 weeks reduced peak infections by approximately 48.88% compared to baseline.

School closure greatly reduced the number of peak infections, but in many cases, it prolonged when the peak day occurred. In fact, it did not have huge results resulting the total number of infections to begin with, as once the school closure ends, people are once again exposed to each other. Additionally, since not everyone will be completely isolated from each other, the transmission rate is only reduced to 20% of its original value, before going back up after the school closure is lifted.

Intervention Comparison

Comparing the three intervention strategies shows that reducing transmission rate is the more effective to lower the number of peak infections. Mask mandates consistently reduced both the peak number of infections and the total epidemic size, especially when using masks that offer higher degrees of protection. The vaccination strategy was also effective, particularly earlier in the outbreak because it reduced the susceptible population while the disease was still growing. School closures delayed the epidemic peak but did not significantly reduce total infections because transmission resumed once the closure ended.

Overall, the results from our simulations suggest strategies focused on reducing overall transmission rates are more effective than temporary interventions, such as short-term closures.

Verify and validate your analysis:

To verify the accuracy and correctness of the model we used, several checks were performed. First, the SEIR model equations were reviewed to ensure that the numerical implementation in Python matched the standard epidemiological SEIR differential equations. The Euler method implementation was verified by confirming that the total population $S+E+I+R$ remained constant throughout the simulation, which ensures that individuals were only transitioning between compartments rather than being artificially created or lost in the model. Next we verified the grid search process by making sure the sum of squared roots decreased as the algorithm tested different combinations of parameters and that the final parameter set corresponded to the lowest SSE observed. Running the SEIR model using these optimized parameters produced infection curves that closely matched the reported outbreak data, particularly around the peak of the epidemic.

This agreement suggests that the parameter values reasonably capture the transmission dynamics of the mystery disease. An R_0 value of approximately 2.91 indicates that, on average, each infected individual spreads the disease to nearly three other people in a fully susceptible population. This value falls within the range observed for several respiratory viruses documented in epidemiological studies. For example, early estimates of the basic reproduction number for the virus responsible for COVID-19 ranged between approximately 2 and 3.5, most closely matching our estimated R_0 value. However, the genome type, genome, particle size, our symptoms and transmission most all aligns with Rubella. TRubella is

a virus spread by single-stranded RNA and is about 50-70nm in diameter (PubMed). The occasional symptoms such as a GI upset and joint pain are also occasional in Rubella (CDC). Additionally, the symptoms of fever, fatigue, sore throat, and rash match best with Rubella (CDC). Rubella replicates in the nasopharynx and regional lymph nodes, consistent with our mystery respiratory infection. The incubation period of 12-23 days also alligns with our outbreak. While Rubella used to be a worldwide epidemic, the most effective intervention strategy was a vaccine (RCV) invented in 1969 (CDC). However, in the 19 countries where access to the RCV vaccine is limited or has not yet been introduced, the CDC models that an estimated 30,000 cases of Rubella will occur each year.

sources: <https://www.cdc.gov/mmwr/volumes/74/wr/mm7418a3.htm>

<https://www.cdc.gov/rubella/about/index.html>

<https://www.cdc.gov/pinkbook/hcp/table-of-contents/chapter-20-rubella.html>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC88950/>

Conclusions and Ethical Implications:

Using exponential growth fitting and SEIR epidemiological modeling, the spread of the mystery virus was analyzed to estimate key epidemic parameters and evaluate intervention strategies. The optimized SEIR model produced an estimated basic reproduction number of $R_0 = 2.91$. The model predicted the epidemic peak to occur around day 82–83, with approximately 3,300–3,600 active infections, representing roughly 20% of the population infected simultaneously. Based on the transmission rate, incubation period, and reported respiratory symptoms, the mystery disease most closely resembles a respiratory viral infection similar to rubella/German measles. Although the estimated R_0 from our model (2.91) is somewhat lower than typical rubella estimates, the clinical symptoms and respiratory transmission pattern most closely resemble rubella infection. Differences between modeled and reported values may result from vaccination coverage or population immunity reducing effective transmission.

Among the intervention strategies tested, mask mandates and vaccination campaigns were the most effective at reducing peak infections because they directly lower the effective transmission rate or reduce the susceptible population (PubMed). School closures delayed the peak but did not significantly reduce the overall epidemic size once the closure ended (PubMed). These results highlight important ethical considerations in public health decision-making. While interventions such as mask mandates or vaccination campaigns can significantly reduce disease spread, they also raise questions about personal autonomy, compliance, and equitable access to healthcare resources. For example, vaccine availability and healthcare access may differ across communities, and restrictive policies such as school closures can disproportionately affect students who rely on campus resources. Additionally, the model assumes full compliance and equal access to interventions, which is not always

realistic. In practice, variations in behavior and resource availability can lead to unequal outcomes across populations, particularly in population dense areas with larger socioeconomic disparities. Policymakers must therefore balance the benefits of disease control with the social and economic impacts of these interventions (PubMed).

Additionally, the model assumes full compliance and equal access to interventions, which is not always realistic. In practice, variations in behavior and resource availability can lead to unequal outcomes across populations. Policymakers must therefore balance the benefits of disease control with the social and economic impacts of these interventions while also recognizing the uncertainty and limitations inherent in model-based predictions (PubMed).

sources: <https://www.cdc.gov/rubella/about/index.html>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8892938/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8909310/>

Limitations and Future Work:

While the SEIR model provided useful insights into the spread of the mystery virus, several limitations should be considered when interpreting the results. First, the model assumes that the transmission rate (β), incubation rate (σ), and recovery rate (γ) remain constant throughout the outbreak. In reality, these parameters often change over time due to behavioral changes, public health interventions, seasonal factors, or viral mutations. Because the model assumes fixed values for these parameters, it may not fully capture the dynamic nature of real epidemics.

Second, the model assumes homogeneous mixing within the population, meaning that every individual has an equal probability of interacting with every other individual. In real populations, contact patterns are highly structured. People interact more frequently within social groups such as households, classrooms, or workplaces, which can significantly affect disease transmission dynamics.

Third, the model assumes perfect compliance with intervention strategies. For example, the mask mandate simulation assumes that all individuals consistently wear masks that reduce transmission by a fixed percentage. In practice, compliance is often incomplete, and mask effectiveness varies depending on proper usage and mask quality.

An additional limitation is that the model does not include additional epidemiological factors such as asymptomatic transmission, reinfection, hospitalization, or mortality. These factors can influence both the spread and severity of an outbreak but were beyond the scope of this simplified SEIR framework.

Future work could improve the accuracy and realism of this model in several ways. One improvement would be to incorporate time-varying transmission parameters, allowing the

transmission rate to change as individuals adopt behavioral changes or as new interventions are implemented. Additionally, the model could be expanded to include age-structured or network-based population models that better represent how individuals interact within communities. Such models would allow researchers to simulate transmission within specific environments such as classrooms, dormitories, or workplaces.

Another potential improvement would be the use of stochastic epidemic models, which incorporate randomness into transmission events (ScienceDirect). These models can be particularly useful during the early stages of an outbreak when case numbers are small and chance events can strongly influence the trajectory of the epidemic.

Finally, integrating real-time surveillance data and vaccination coverage information could allow the model to support real-world public health decision-making. By continuously updating model parameters as new data becomes available, epidemiological models could provide more accurate forecasts and guide effective intervention strategies during future outbreaks.

Sources: <https://www.sciencedirect.com/science/article/pii/S2468042716300495>